Supplemental Online Content

Griffiths CD, Zhang B, Tywonek K, Meyers BM, Serrano PE. Toxicity profiles of systemic therapies for advanced hepatocellular carcinoma: a systematic review and meta-analysis. *JAMA Netw Open.* 2022;5(7):e2222721. doi:10.1001/jamanetworkopen.2022.22721

eFigure 1. Liver-Related Toxic Effects in Placebo-Controlled Trials and Trials With Active Comparator

eFigure 2. Serious Adverse Events in Placebo-Controlled Trials and in Trials With Active Comparator

eFigure 3. Grade 3 or Higher Adverse Events in Placebo-Controlled Trials and Trials With Active Comparator

eFigure 4. Risk of Bias 2 Assessment

eTable. MINORS Assessment

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This supplemental material has been provided by the authors to give readers additional information about their work.

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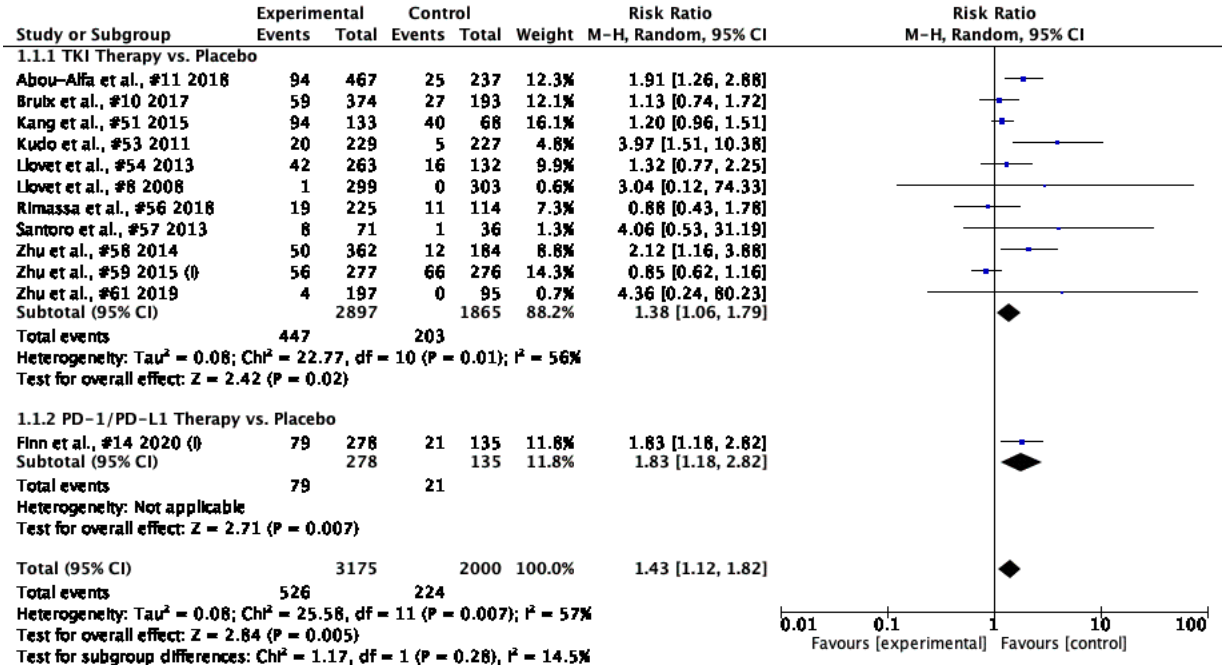
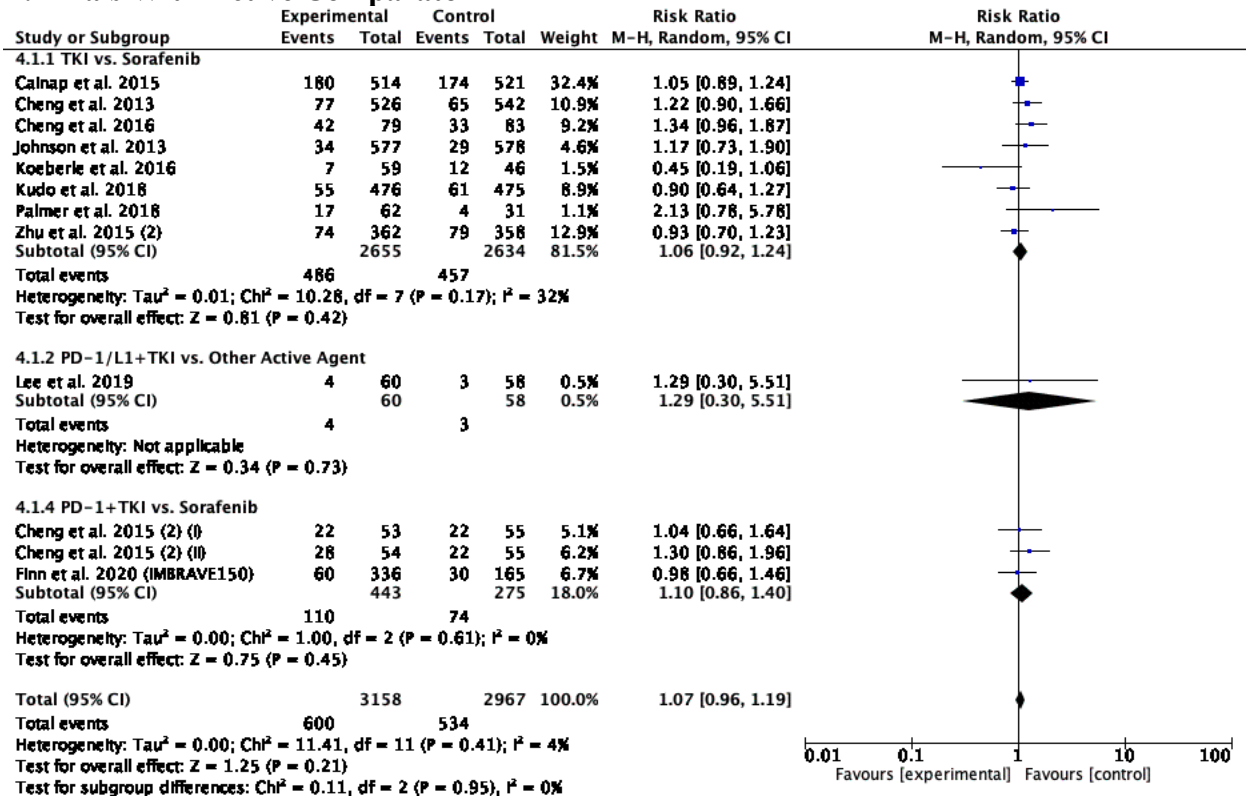
eFigure 1. Liver-Related Toxic Effects in Placebo-Controlled Trials and Trials With Active Comparator

A. Placebo-Controlled Trials



B. Trials With Active Comparator



Lee et al., 2019 compares atezolizumab + bevacizumab to atezolizumab. The remaining comparators are Sorafenib.

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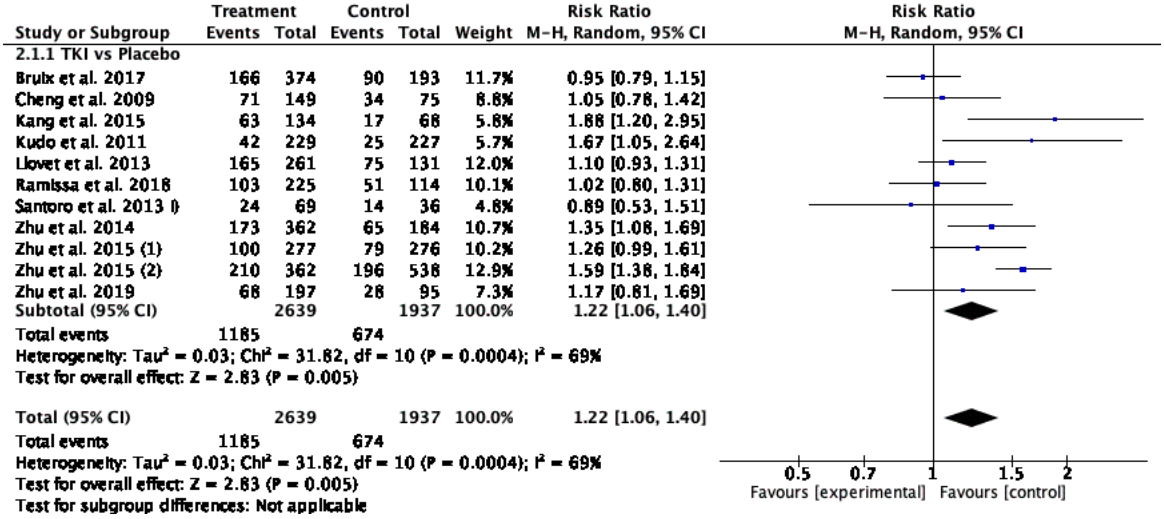
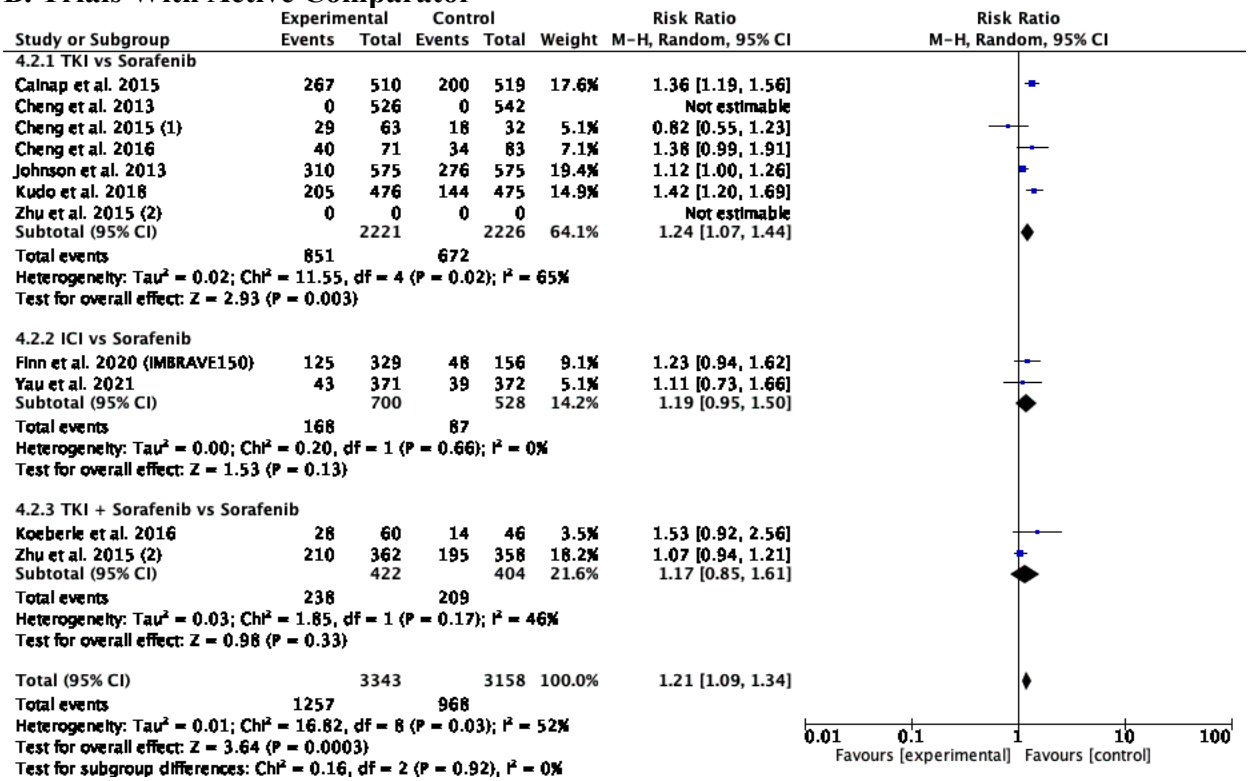
eFigure 2. Serious Adverse Events in Placebo Controlled Trials and in Trials With Active Comparator

A. Placebo-Controlled Trials



B. Trials With Active Comparator



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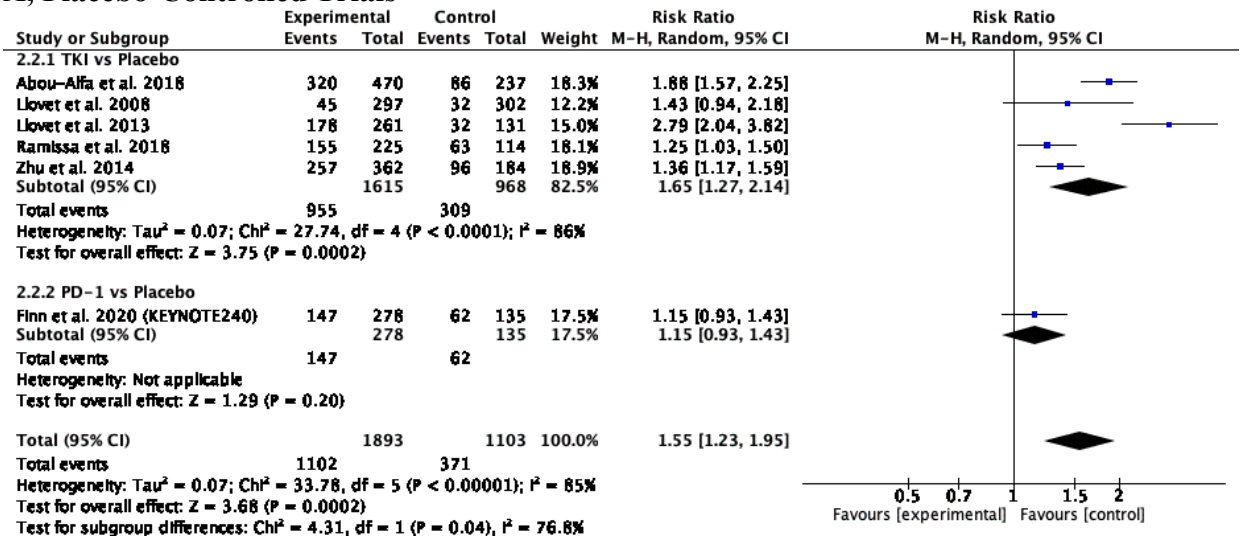
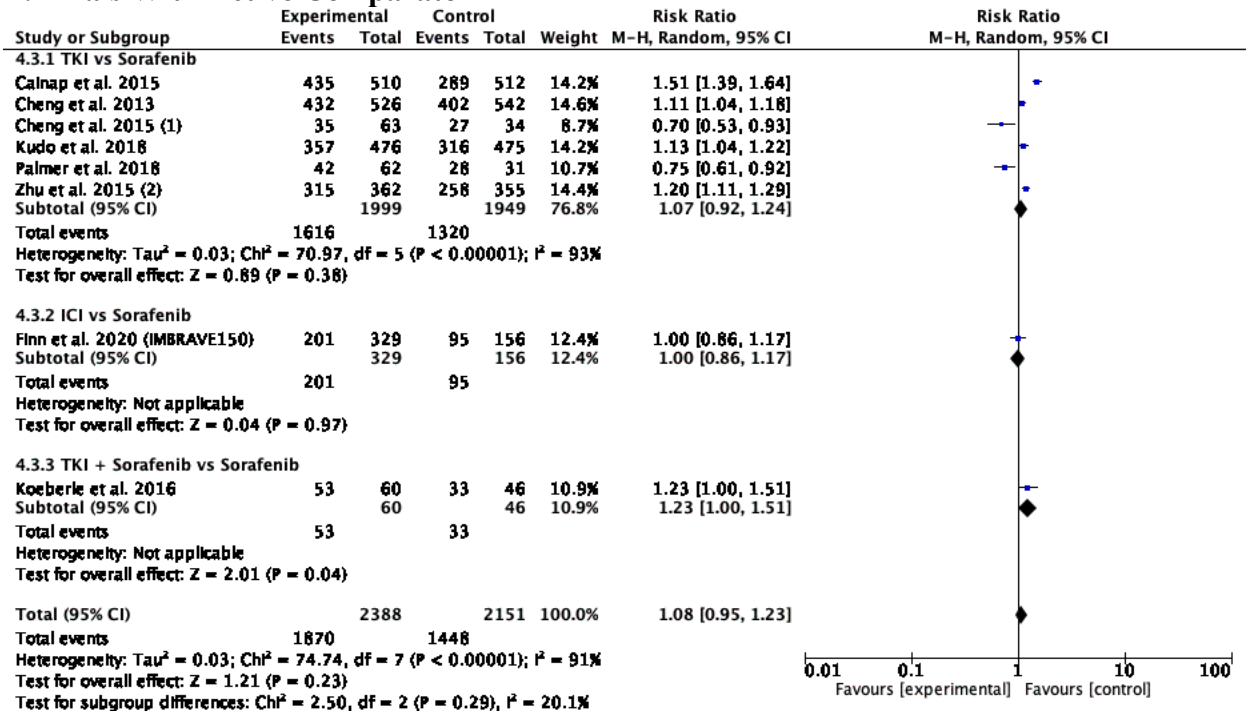
eFigure 3. Grade 3 or Higher Adverse Events in Placebo-Controlled Trials and Trials With Active Comparator

A, Placebo-Controlled Trials



B. Trials With Active Comparator



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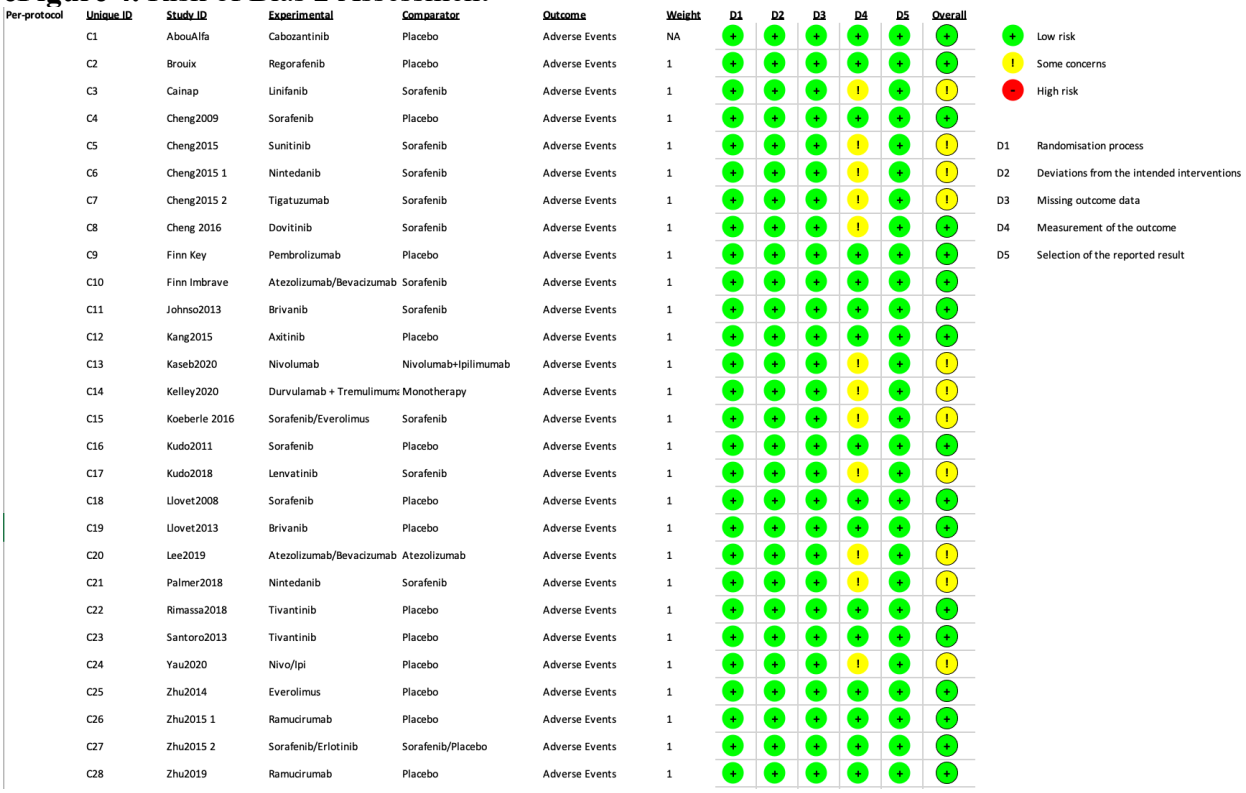
eFigure 4. Risk of Bias 2 Assessment



Per-protocol	Unique ID	Study ID	Experimental	Comparator	Outcome	Weight	D1	D2	D3	D4	D5	Overall
	C1	AbouAlfa	Cabozantinib	Placebo	Adverse Events	NA	+	+	+	+	+	+
	C2	Brouix	Regorafenib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C3	Cainap	Unifanib	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C4	Cheng2009	Sorafenib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C5	Cheng2015	Sunitinib	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C6	Cheng2015 1	Nintedanib	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C7	Cheng2015 2	Tigatuzumab	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C8	Cheng 2016	Dovitinib	Sorafenib	Adverse Events	1	+	+	+	!	+	+
	C9	Finn Key	Pembrolizumab	Placebo	Adverse Events	1	+	+	+	+	+	+
	C10	Finn Imbrave	Atezolizumab/Bevacizumab	Sorafenib	Adverse Events	1	+	+	+	+	+	+
	C11	Johnso2013	Brivanib	Sorafenib	Adverse Events	1	+	+	+	+	+	+
	C12	Kang2015	Axitinib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C13	Kaseb2020	Nivolumab	Nivolumab+Ipilimumab	Adverse Events	1	+	+	+	!	+	!
	C14	Kelley2020	Durvulamab + Tremulimum	Monotherapy	Adverse Events	1	+	+	+	!	+	!
	C15	Koeberle 2016	Sorafenib/Everolimus	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C16	Kudo2011	Sorafenib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C17	Kudo2018	Lenvatinib	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C18	Llovet2008	Sorafenib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C19	Llovet2013	Brivanib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C20	Lee2019	Atezolizumab/Bevacizumab	Atezolizumab	Adverse Events	1	+	+	+	!	+	!
	C21	Palmer2018	Nintedanib	Sorafenib	Adverse Events	1	+	+	+	!	+	!
	C22	Rimassa2018	Tivantinib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C23	Santoro2013	Tivantinib	Placebo	Adverse Events	1	+	+	+	+	+	+
	C24	Yau2020	Nivo/Ipi	Placebo	Adverse Events	1	+	+	+	!	+	!
	C25	Zhu2014	Everolimus	Placebo	Adverse Events	1	+	+	+	+	+	+
	C26	Zhu2015 1	Ramucirumab	Placebo	Adverse Events	1	+	+	+	+	+	+
	C27	Zhu2015 2	Sorafenib/Erlotinib	Sorafenib/Placebo	Adverse Events	1	+	+	+	+	+	+
	C28	Zhu2019	Ramucirumab	Placebo	Adverse Events	1	+	+	+	+	+	+

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eTable. MINORS Assessment

	Marron et al., 2022	Finn et al., 2020 (iii)
A clearly stated aim	2	2
Inclusion of consecutive patients	0	2
Prospective collection of data	2	2
Appropriate endpoints	2	2
Unbiased assessment of endpoints	2	2
Appropriate follow up	2	2
Loss to follow up <5%	2	2
Prospective calculation of sample size	2	2
Total	14/16	16/16

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eFigure 5. GRADE of Evidence Assessment

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	TKIs	ICIs	Relative (95% CI)	Absolute (95% CI)		
IV Liver												
23	randomised trials	not serious	not serious	serious[a]	not serious	none	1688/8040 (21.0%)	301/1075 (28.0%)	**Proportion 0.23** (0.18 to 0.28)	**-- per 1,000** (from -- to --)	⨁⨁⨁◯ Moderate	
IV serious AEs												
23	randomised trials	not serious	not serious	serious	not serious	none	2998/6517 (46.0%)	304/1266 (24.0%)	**Proportion 0.41** (0.34 to 0.48)	**-- per 1,000** (from -- to --)	⨁⨁⨁◯ Moderate	
IV grade 3<												
20	randomised trials	not serious	not serious	serious[a]	not serious	none	3908/5665 (69.0%)	555/1587 (35.0%)	**Proportion 0.56** (0.46 to 0.67)	**-- per 1,000** (from -- to --)	⨁⨁⨁◯ Moderate	

CI: confidence interval

Explanations

a. Proportions largely compared between different studies

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